Custom Instruction CRC Design on Nios® V/g Processor Agilex™ 7 FPGA

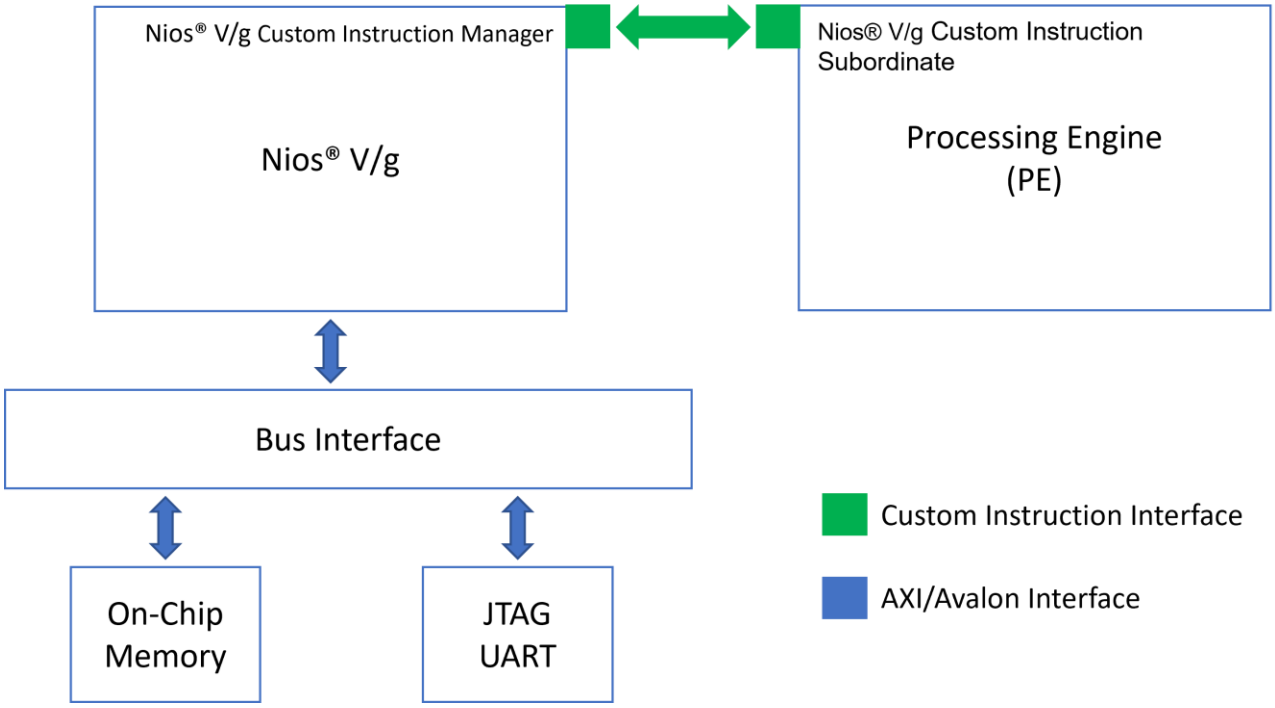
Contents

1. Theory of Operation

This design demonstrates the custom instruction feature of the Nios® V/g processor using Intel Agilex® 7 F-Series FPGA Development Kit.

1.1. Block Diagram

A Processing Engine (PE) that performs basic arithmetic and logical computations is connected to the Nios® V/g processor using the custom instruction interface. The current version of the Nios® V/g processor custom instruction interface supports operations up-to 32-Bit.



1.2. IP Cores

The following IPs are used in this design.

- Nios V/g soft processor core
- On Chip RAM-II
- JTAG UART
- Custom PE 1
- System ID

2. Directory Structure

The directory structure is explained below:

Root directory: `agf014ea-dev-devkit/niosv_g/ci_crc`

Directories	Content
docs	Document capturing the details of the example design
img	Block diagram for the example designs
ready_to_test	Prebuilt binaries (FPGA `.sof` file and Application `.elf` file) which can be programmed on the board and tested
sources - hw - scripts - sw/app	Files needed to create the design - Necessary hardware files (`.sv`, `.v`, `.ip`) of the design - This folder consists of scripts to build the design - This folder contains software application files

3. Quick Start Guide

You may try the example on the target development kit using the prebuilt binaries. The respective FPGA `sof` and Application `elf` files are found in `ready_to_test` folder.
Refer to Chapter 6 Programming the Design for the steps to program these files.

4. Generating the Design

The implementation of Nios® V processor system requires a hardware design developed using the Quartus® Prime software. After that, it is required to develop the software design that complements the processor system.

4.1. Building Hardware Design

Begin designing the system hardware by instantiating the Nios® V processor and its peripherals into Platform Designer. After configuring the system assignments and constraints, complete the hardware design by performing a successful compilation.

We will be using `sources/scripts/build_sof.py` to develop the hardware design.

This script will create a new platform design, add the IPs into the design, makes the connections, compiles the design and generates the .sof file.

Steps to execute the script are as follows:

1. Invoke the Nios V Command Shell in the terminal
2. Run the following command in the terminal from top level project directory: `> quartus_py ./scripts/build_sof.py`
3. The Quartus tool will compile the design and generate the output files into `<Root Directory>/sources/hw/output_files`

4.2. Building Software Design

After the processor system is ready, you may begin building the software design. It consists of the following steps:

1. Create a board support package (BSP) project.
2. Create a Nios® V processor application project with source code.
3. Build the application.

4. Convert the application .elf file into .hex file for memory initialization.

To create software application, run the following commands in the terminal:

```
> niosv-bsp -c --quartus-project=hw/ci_crc.qpf --qsys=hw/sys.qsys --type=hal
sw/bsp_crc/settings.bsp

> niosv-app --bsp-dir=sw/bsp_crc --app-dir=sw/app_crc --srcs=sw/app_crc/srcs/

> niosv-shell

> cmake -S ./sw/app_crc -G "Unix Makefiles" -B sw/app_crc/build

> make -C sw/app_crc/build

> niosv-download -g sw/app_crc/build/app_crc.elf -c 1
```

Note: It is recommended to clean the `app/build` project before regenerating `elf` (cmake and make).

5. Simulating the Design

You may carry on simulation for this design to evaluate its intended functions. During simulation, the action of downloading .elf file is unable to be simulated. Thus, the processor memory is instead initialized with the application .hex file.

Use the following commands to run the simulation:

```
# Generate Testbench from Platform Designer.

# Generate -> Generate Testbench System

> cp ./hw/onchip_mem.hex ./sys_tb/sys_tb/sim/mentor

> cd hw/sys_tb/sys_tb/sim/mentor/

> vsim &

> source msim_setup.tcl

> ld_debug

> run -all
```

6. Programming the Design

Use the Quartus® Prime Programmer tool and the Ashling* RiscFree* IDE to program the Nios® V processor-based system into the FPGA and to run your application.

Program the generated `sof`, then download the `elf` file on the board according to the command below.

```
> quartus_pgm --cable=1 -m jtag -o 'p;ready_to_test/ci_crc.sof'

#----- Download the elf file on the board ----#

> niosv-download -g ready_to_test/app_crc.elf -c 1

#----- Verify the output on the terminal by using the following command in the terminal -----#

> juart-terminal -c 1 -i 0
```

Note: Reduce the JTAG clock frequency to 6MHz before programming the application `.elf` file on the board using the command: `jtagconfig --setparam 1 JtagClock 6M`

6. Expected Results

The following is the output as observed on the JTAG UART terminal. The output is analogous to the logic from the application code. Users should be able to observe the same output on their terminal/setup.

Figure 1-2. Expected Result

```
+------------------------------------------------------------++
| Comparison between software and custom instruction CRC32   |
+------------------------------------------------------------+


System specification
--------------------
System clock speed = 50 MHz
Number of buffer locations = 32
Size of each buffer = 256 bytes


Initializing all of the buffers with pseudo-random data
-------------------------------------------------------
Initialization completed


Running the software CRC
------------------------
Completed


Running the optimized software CRC
----------------------------------
Completed


Running the custom instruction CRC
----------------------------------
Completed


Validating the CRC results from all implementations
---------------------------------------------------
All CRC implementations produced the same results


Processing time for each implementation
---------------------------------------
Software CRC = 58 ms
Optimized software CRC = 39 ms
Custom instruction CRC = 04 ms


Processing throughput for each implementation
---------------------------------------------
Software CRC = 1129 Kbps
Optimized software CRC = 1680 Kbps
Custom instruction CRC = 16384 Kbps


Speedup ratio
-------------
Custom instruction CRC vs software CRC = 14
Custom instruction CRC vs optimized software CRC = 9
Optimized software CRC vs software CRC= 1
Print the value of System ID
System ID from Peripheral core is 0xA6B7C8D9
```